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The meshless numerical simulation of Kelvin–Helmholtz instability during the wave growth of liquid–liquid slug flow

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ABSTRACT

Kelvin–Helmholtz instability (KHI) occurs at the interface of two fluids, in which the heavier fluid flows at the bottom. In the present numerical work, the KHI was analyzed by solving the modification of two-dimensional (2-D) incompressible Navier–Stokes equations. To capture the interface, the Cahn–Hilliard equation was implemented into the Navier–Stokes equations. The phenomena around the KHI of two and three-component fluids were investigated numerically by using radial basis function (RBF) combined with the domain decomposition method (DDM) in a primitive variable formulation. Here DDM is able to solve the large scale problem. On the other hand the calculation accuracy decreases with the increase of the number of the subdomains. For the above reason, in the present works, the domain was partitioned into 15 x 15 subdomains in order to reduce the decrease in accuracy due to the domain division. Next, fractional step method was used to solve the modification of the Navier–Stokes equations.

The numerical results indicate that the procedure used in the present work can easily handle the KHI problem under the variations of the interface thickness, density ratios, and initial velocity differences. Moreover, the interface evolutions obtained from the present method agree well with those of the finite difference method. The effects of the interface thickness, density ratio and magnitude of velocity difference on the KHI were also investigated. The decrease of the interface thickness produces a non-smooth concentration profile, and the increase of the interface thickness produces too much surface diffusion. It was found also that the increase of the density ratio reduces the growth of KHI. The interface rolls up are strongly affected by the initial horizontal velocity difference. Finally, the present study also shows that the RBF method is a reliable method to solve the KHI on the complex domains.

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1. Introduction

Slug flows are often encountered in the practical applications during the hydrocarbons transportation pipeline system [1]. In general, it is intermittent and is formed by sequences of large gas bubbles followed by the liquid slug. It flows under the high frequencies in which large gas bubbles flow alternately with liquid slug [2,3]. In the horizontal pipes, it is formed from a stratified flow due to the growth of hydrodynamic instabilities. Here small perturbations in the form of small waves naturally emerge. Those waves continue to grow, capturing the liquid that flows in front of them until the pipe cross-sectional area is fully blocked by the liquid, thus forming the slugs [4]. This wave growth mechanism is known as the Kelvin–Helmholtz instability (KHI) [5,6].

Slug flow is one of the dangerous flow patterns in the petroleum industry. In order to ensure the safe operation of the pipeline, the knowledge of the slug flow mechanism is essential. The appearance of slug flow in the pipeline will increase the structural disadvantage that leads to crack and corrosion of the pipe [7], whereas high-frequency slug flows increase the rate of the pipeline corrosion [8]. Consequently, it will decrease the hydrocarbon production and increase the equipment maintenance cost [9]. From the viewpoint of multiphase-flow, the understanding of KHI is very important in pipeline design.

The researchers have developed the numerical methods to simulate the KHI in horizontal pipes or horizontal channel. Simmons et al. [10] explored the influence of KHI during the transition to slug flow in horizontal pipes. The basic mechanism of KHI is explained as the existence of the uniform shear stress between two different fluids. To simplify the calculation, the surface tension and viscosity were neglected, and, consequently, KHI is considered only as the fluid instability of a lighter fluid is placed on the top of a heavier one. San & Kara [11] calculated 2-D KHI in order to examine the performance of six Riemann flux formulas. Those formulas are the combination of the Rusanov scheme, Reo scheme, Harten–Lax–Van Leer scheme, first-order centered scheme, advection upstream splitting method, and Marquina scheme. Here, a single mode perturbation was used as a tested case. They concluded that the advection upstream splitting scheme results in the smallest numerical dissipation.

Patnaik et al. [12] developed a numerical simulation for KHI between two stratified fluids by using the finite difference method. They concluded that the minimum Richardson number required for the onset of KHI was at Reynolds number of 100. They claimed that their result is in a good agreement with that of the linear stability theory. On the other hand, Balabel et al. [13] introduced the level set method to solve the hydrodynamic instability. The numerical procedure was developed on the basis of the solution of Navier–Stokes equations and structural 2-D computational grid by using the control volume approach. The results showed that the onset of roll-up mechanism at the interface is largely affected by the considered flow regimes. Next, Lee et al. [14] examined the finite difference method to solve the KHI of multi-component fluids. It was modeled by both of modification of Navier–Stokes and the multi-component Cahn–Hilliard equations. They concluded that the linear growth rate for the KHI of two-component fluids was in good agreement with the analytical results. These finite difference and finite volume methods used the mesh to solve problems, and it can produce lot of numerical problems during the discretizing of complex domains with irregular boundaries [15]. Moreover, those methods were developed on the basis of the polynomial interpolations, whereas they are limited by the algebraic convergence rates [16].

Shadloo et al. [17] used the smoothed particle hydrodynamics (SPH) scheme to simulate the 2-D KHI problem of an incompressible two-phase immiscible fluid. They concluded that the numerical algorithm is able to handle the two-phase fluid flow under the variation of the density ratios. Yue et al. [18] studied also the growth of the KHI at the interface between two ideal gases using the SPH scheme. The linear growth rate in the numerical solution was found to be in a satisfactory agreement with that of the analytical solution, in which the average relative error was 13%. As the SPH is an approximate method, it is noticed that there are some disadvantages if it is not applied correctly in a specific problem. First, it is very sensitive to the number of particles used in the spatial approximation. Second, the precision of the SPH depends on the order of accuracy of the window function [19].

The above facts indicate that there is a strong disagreement from the previous works regarding the best solution to describe the physical phenomena around KHI. For this reason, a systematic numerical study to explain the involved phenomena around the transition to slug flow of liquid–liquid two-phase flow is needed. The aim of the present study is to carry out a numerical investigation of the KHI during the transition to slug flow. In the present numerical study, the radial basis function (RBF) method is used to simulate the phenomena using modification of the Navier–Stokes equations in terms of primitive variables and the Cahn–Hilliard equations in order to capture the evolution of the interface between two fluids.

An attractive and simple approach that is suitable for a wide range of the problems employs the capability of radial basis function (RBF) [20]. It has a good capability to solve the governing equations in the strong form, without any integration is required. In addition, it is also effective to solve the numerical terms with a low numerical diffusion [21]. The main challenge to implement the RBF method is the applicability to handle the large-scale domain. RBF method should be combined with the DDM, when handling large-scale domain. The basic idea of DDM is to divide one large domain into a number of smaller subdomains [22]. The DDM is able to handle large-scale problems, meanwhile it will decrease the calculation accuracy if more subdomains are employed [23].

Several numerical experiments were carried-out in order to examine the accuracy of RBF method combined with DDM. Mai-Duy et al. [22] used it to solve one and two-dimensional elliptic problems. The domain was divided into sixteen subdomains, and the results show a high level of accuracy and a fast rate of convergence. Therefore it presents more attractive than the one-domain method to solve the elliptic problems under a large computational domain. Zhou et al. [23] developed RBF method combine with DDM to solve the elliptic partial differential equations. They investigated the accuracy of the numerical solution during the increase of the numbers of subdomains. In case of 16 × 16 subdomains, they found that the root-mean square error (RMSE) was 0.005972, which represents a high accuracy of the numerical method.

In general, the RBF and SPH methods have several similarities. Those are as follows: (1) they are simple to implement, (2) they are a truly meshless method, and (3) they were developed on the basis of strong form of partial differential equations. Moreover, the RBF method has a better accuracy than the SPH method at the expense of more computational effort [24]. The need to invert the coefficient matrix makes the RBF more expensive than SPH method. However, the efficiency of the RBF method can be improved by using the domain decomposition method.

In the present numerical works, the combination of RBF and DDM was implemented, because it is a very efficient method and high accuracy with low numerical diffusion as suggested by [15] and [25]. The DDM is used to solve the multi small subdomain problems instead of single large domain problem. Next, the Cahn–Hilliard equation is used to capture the evolution of the interface, because it is a thermodynamically consistent equation in order to follow the interface evolution as reported by [26] and [27].

2. Radial Basis Function (RBF) method

The main advantage of RBF approximation is the ability to work for high dimensions of the arbitrary geometries. RBF is a function whose value depends only on the distance from some center point. By using the distance functions, it can be easily implemented to reconstruct a plane or surface using scattered data in two- and three-dimensional (2-D and 3-D) or higher dimensional spaces. It was initially developed for multivariate data and function interpolation. Hence, the meshfree nature has motivated researchers to employ them in solving partial differential equations.

In the RBF method, the approximation of the function f is written as the linear combination of the RBF, given by [28]

$$f = \sum_{j=1}^{N} \alpha_j \varphi \tag{1}$$

The basic definition of the basis function φ is

$$\varphi_{i,j} = \sqrt{r_{i,j} + \varepsilon^2} \qquad i, j = 1, 2, \ldots, N \tag{2}$$

where ε is a shape parameter. The first and second derivatives are given as

$$\frac{\partial \varphi}{\partial x} = \frac{x_i - x_j}{\sqrt{r_{i,j}^2 + \varepsilon^2}}, \quad \frac{\partial \varphi}{\partial y} = \frac{y_i - y_j}{\sqrt{r_{i,j}^2 + \varepsilon^2}}, \quad \frac{\partial^2 \varphi}{\partial x^2} = \frac{\left(y_i - y_j\right)^2 + \varepsilon^2}{\left(r_{i,j}^2 + \varepsilon^2\right)^{3/2}}, \quad \frac{\partial^2 \varphi}{\partial y^2} = \frac{\left(x_i - x_j\right)^2 + \varepsilon^2}{\left(r_j^2 + \varepsilon^2\right)^{3/2}} \tag{3}$$

$$r_{i,j} = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2} \qquad i, j = 1, 2, \ldots, N \tag{4}$$

where α is the expansion coefficient, x and y are the spatial vectors, r is the distance matrix, ε is the shape parameter and φ is the radial basis function. The expansion coefficients α are selected by enforcing the interpolation condition at the set of nodes that typically coincide with the centers. The interpolation condition at N centers results in a NxN linear system

$$B\alpha = f \tag{5}$$

to be solved for RBF expansion coefficient α. The matrix B is the interpolation matrix with entries:

$$b_{i,j} = \varphi_{i,j} = \sqrt{r_{i,j}^2 + \varepsilon^2} \tag{6}$$

$$B = \begin{bmatrix} b_{1,1} & b_{1,2} & \cdots & b_{1,N} \\ b_{2,1} & b_{2,2} & \cdots & b_{2,N} \\ b_{3,1} & b_{3,2} & \cdots & b_{3,N} \\ \varphi_{\cdot 1,1} & \varphi_{\cdot 1,1} & \cdots & \varphi_{\cdot 1,1} \\ \varphi_{\cdot 1,1} & \varphi_{\cdot 1,1} & \cdots & \varphi_{\cdot 1,1} \\ \varphi_{\cdot 1,1} & \varphi_{\cdot 1,1} & \cdots & \varphi_{\cdot 1,1} \\ b_{N,1} & b_{N,2} & \cdots & b_{N,N} \end{bmatrix} \tag{7}$$

The expansion coefficient α can be defined as

$$\alpha = \begin{bmatrix} \alpha_1 & \alpha_2 & \cdots & \alpha_N \end{bmatrix}^T \tag{8}$$

$$f = \begin{bmatrix} f(\mathbf{x}_1) & f(\mathbf{x}_2) & \cdots & f(\mathbf{x}_N) \end{bmatrix}^T \tag{9}$$

$$\alpha = B^{-1}f \tag{10}$$

The approximation of the first derivative $\left(\frac{\partial f}{\partial x}\right)$ is expressed as

$$\frac{\partial f}{\partial x} = \sum_{j=1}^{N} \alpha_j \frac{\partial \varphi}{\partial x} \tag{11}$$

$$\frac{\partial f}{\partial x} = H_x \alpha \tag{12}$$

H_x is the $N \times N$ evaluation matrix with entries:

$$hx_{i,j} = \left(\frac{\partial \varphi}{\partial x}\right)_{i,j} = \frac{x_i - x_j}{\sqrt{r_{i,j}^2 + \varepsilon^2}} \tag{13}$$

$$H_x = \begin{bmatrix} hx_{1,1} & hx_{1,2} & \cdots & hx_{1,N} \\ hx_{2,1} & hx_{2,2} & \cdots & hx_{2,N} \\ hx_{3,1} & hx_{3,2} & \cdots & hx_{3,N} \\ \varphi\cdot_{1,1} & \varphi\cdot_{1,1} & \cdots & \varphi\cdot_{1,1} \\ \varphi\cdot_{1,1} & \varphi\cdot_{1,1} & \cdots & \varphi\cdot_{1,1} \\ \varphi\cdot_{1,1} & \varphi\cdot_{1,1} & \cdots & \varphi\cdot_{1,1} \\ hx_{N,1} & hx_{N,2} & \cdots & hx_{N,N} \end{bmatrix} \tag{14}$$

By substituting $\alpha = B^{-1}f$ into Eq. (11), the approximation of the first derivative is defined as

$$\frac{\partial f}{\partial x} = H_x B^{-1} f \tag{15}$$

The differentiation matrix D_x can be written as

$$D_x = H_x B^{-1} \tag{16}$$

$$\frac{\partial f}{\partial x} = D_x f \tag{17}$$

The second derivative $\frac{\partial^2 f}{\partial x^2}$ is approximated as

$$\frac{\partial^2 f}{\partial x^2} = \sum_{j=1}^{N} \alpha_j \frac{\partial^2 \varphi}{\partial x^2} \tag{18}$$

$$\frac{\partial^2 f}{\partial x^2} = H_{xx} \alpha \tag{19}$$

H_{xx} is the $N \times N$ evaluation matrix with entries:

$$hxx_{i,j} = \left(\frac{\partial^2 \varphi}{\partial x^2}\right)_{i,j} = \frac{\left(y_i - y_j\right)^2 + \varepsilon^2}{\left(r_{i,j}^2 + \varepsilon^2\right)^{3/2}} \tag{20}$$

$$H_{xx} = \begin{bmatrix} hxx_{1,1} & hxx_{1,2} & \cdots & hxx_{1,N} \\ hxx_{2,1} & hxx_{2,2} & \cdots & hxx_{2,N} \\ hxx_{3,1} & hxx_{3,2} & \cdots & hxx_{3,N} \\ \varphi\cdot_{1,1} & \varphi\cdot_{1,1} & \cdots & \varphi\cdot_{1,1} \\ \varphi\cdot_{1,1} & \varphi\cdot_{1,1} & \cdots & \varphi\cdot_{1,1} \\ \varphi\cdot_{1,1} & \varphi\cdot_{1,1} & \cdots & \varphi\cdot_{1,1} \\ hxx_{N,1} & hxx_{N,2} & \cdots & hxx_{N,N} \end{bmatrix} \tag{21}$$

$$\frac{\partial^2 f}{\partial x^2} = H_{xx} B^{-1} f \tag{22}$$

The differentiation matrix D_{xx} is calculated as

$$D_{xx} = H_{xx} B^{-1} \tag{23}$$

$$\frac{\partial^2 f}{\partial x^2} = D_{xx} f \tag{24}$$

Moreover, $\frac{\partial f}{\partial y}$ and $\frac{\partial^2 f}{\partial y^2}$ are calculated by using the same procedure.

The advantage of RBF method over the classical method such as, finite difference and finite element method, is the meshless property of these methods [29]. The comparison of RBF method with the finite element method has been carried-out by [30]. The result revealed the superior accuracy of the RBF method in comparison to that of the finite element

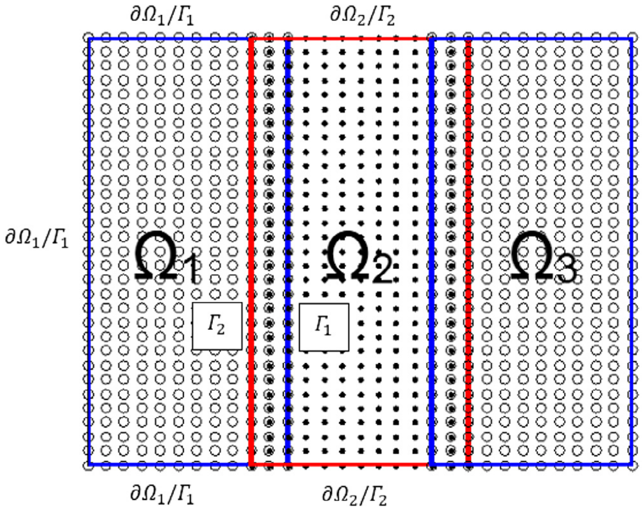

Fig. 1. The illustration of the domain decomposition method.

method. Theoretically, the RBF method is exponentially or spectrally accurate and is completely grid free [31,32]. The RBF method has a spectral convergence rate and has an error curve that continues down as the error plot of the spectral method. The error decays at the rate $O\left(\gamma^N\right)$ where $0 < \gamma < 1$, and N is the number of node [16].

The accuracy of RBF can be improved by increasing the number of N or by increasing the magnitude of the shape parameter [30]. While increasing N leads to the heavy computations, and increasing the shape parameter, as the error becomes smaller, but the matrix becomes more ill-conditioned, hence the solution will break down. Nevertheless, there exists a wide range of shape parameters that the best approximation results can be produced [32]. The technique to obtain an optimal value of the shape parameter has been developed in [33,34].

Meanwhile the radial basis function method may face the numerical difficulty during the solving of large-scale domain. For this reason, it works well if the domain decomposition method is implemented.

2.1. Domain decomposition method

The basic idea of the domain decomposition method is to partition the large computational domain (Ω) into a number of subdomains (Ω_i). Fig. 1 shows the illustration of the domain decomposition method used in the present numerical work. As shown in the figure, the computational domain is divided into three subdomains (Ω_1, Ω_2, and Ω_3), where $\Omega_1 \cap \Omega_2 \neq 0$, and $\Omega_2 \cap \Omega_3 \neq 0$. In the figure, the artificial boundary Γ_i is the boundary of Ω_i that interior of Ω and the rest of boundaries are defined by $\partial\Omega_i \backslash \Gamma_i$. As an illustration, the elliptic partial differential equation (PDE) on the subdomain (Ω_1, Ω_2) can be solved by Schwarz algorithm, which is given as [35]:

$$\mathcal{L}U_1^n = f(x), \qquad \text{in } \Omega_1, \tag{25}$$

$$\mathcal{B}U_1^n = g(x), \qquad \text{on } \partial\Omega_1|\Gamma_1, \tag{26}$$

$$U_1^n = U_2^{n-1}|_{\Gamma_i}, \qquad \text{on } \Gamma_1, \tag{27}$$

and

$$\mathcal{L}U_2^n = f(x), \qquad \text{in } \Omega_2, \tag{28}$$

$$\mathcal{B}U_2^n = g(x), \qquad \text{on } \partial\Omega_2|\Gamma_2, \tag{29}$$

$$U_2^n = U_1^n|_{\Gamma_2}, \qquad \text{on } \Gamma_2, \tag{30}$$

where U_i^n is the solution of the subdomain Ω_i.

3. Problem definition

For the simulation of KHI, it is considered that the 2-D motion of two immiscible fluids of different densities and velocities separated by an interface, in which the heavier fluid is at the bottom. In the numerical analysis, the Cartesian coordinates were selected. For simplicity, the x-dimension of the computational domain L $(0 < x < L)$ is chosen to be equal to the domain height H $(L = H)$. The fluids move counter-currently at the uniform velocity U as shown in Fig. 2. In the calculation, two-fluid system was perturbed by a sinusoidal disturbance at the interface.

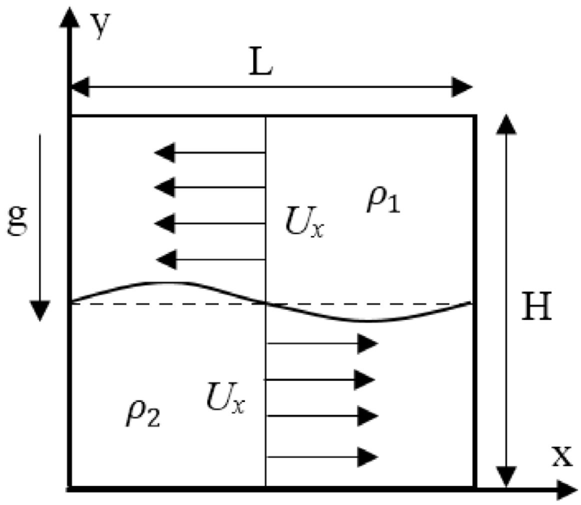

Fig. 2. Schematic diagram of Kelvin–Helmholtz instability (KHI) problem.

3.1. Governing equations

The involved governing equations for the KHI are mathematically expressed by the momentum, mass conservation, and the convective Cahn–Hilliard equations at each point of the computational domain. The governing equations can be written in the following form:

$$\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}=-\frac{1}{\rho(c)}\frac{\partial p}{\partial x}+\frac{\eta}{Re.\rho\,(c)}\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)+SF\,(c) \tag{31}$$

$$\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}=-\frac{1}{\rho(c)}\frac{\partial p}{\partial y}+\frac{\eta}{Re.\rho\,(c)}\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)+SF\,(c)+\frac{g}{Fr^2} \tag{32}$$

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0 \tag{33}$$

$$\frac{\partial c}{\partial t}+u\frac{\partial c}{\partial x}+v\frac{\partial c}{\partial y}=\frac{1}{Pe}\left(\frac{\partial^2 \mu}{\partial x^2}+\frac{\partial^2 \mu}{\partial y^2}\right) \tag{34}$$

$$\mu=f\,(c)-\epsilon^2\left(\frac{\partial^2 c}{\partial x^2}+\frac{\partial^2 c}{\partial y^2}\right)+\beta(c) \tag{35}$$

$$f\,(c)=c\,(c-0.5)\,(c-1) \tag{36}$$

$$\beta\,(c)=-c\,(f\,(c)+f\,(c-1)) \tag{37}$$

$$\rho\,(c)=\rho_1 c+\rho_2\,(c-1) \tag{38}$$

The above equations are obtained using the following dimensionless variables:

$$x'=\frac{x}{L_c},\quad u'=\frac{u}{U_c},\quad t'=\frac{tU_c}{L_c},\quad \rho'=\frac{\rho}{\rho_c}, \tag{39}$$

$$p'=\frac{p}{\rho_c U_c^2},\quad \eta'=\frac{\eta}{\eta_c},\quad g'=\frac{g}{g_c},\quad \mu'=\frac{\mu}{\mu_c} \tag{40}$$

where $'$ represents the dimensionless values.

The parameters of L_c, U_c, ρ_c, η_c, μ_c, SF and g_c are the characteristic quantities of length, velocity, density, viscosity, chemical potential, surface force and acceleration due to gravity respectively. Those are simplified by using the dimensionless groups such as Reynolds number ($Re=\rho_c U_c L_c/\eta_c$), Weber number $\left(We=\rho_c L_c U_c^2/\sigma\right)$, Froude number $\left(Fr=U_c/\sqrt{gL_c}\right)$ and Peclet number ($Pe=U_c L_c/(M\mu)$).

The momentum equations are hydro-dynamically coupled to the Cahn–Hilliard equations. The left-hand side of Eqs. (31) and (32) is the rate of change of momentum, and the right-hand side is a sum of the acting forces on the fluids. Eq. (33) is the mass conservation of the fluid, which assumes that the mass density is uniform throughout the fluid. Eq. (34) is the Cahn–Hilliard equation used in the present numerical study. Eqs. (38) describes the density of the fluid as a function of the concentrations.

In the present study, the initial conditions both of the velocity and concentration are defined by hyperbolic tangent function, as suggested by [14] as follows:

$$l\left(x,y;a,b\right)=tanh\left[\left(y-b-0.01sin\left(a\pi x\right)\right)/\left(2\sqrt{2}\epsilon\right)\right] \tag{41}$$

whereas, the velocity and concentration are considered by the following equations.

$$u\left(x,y,o\right)=l\left(x,y;2,0.5\right) \tag{42}$$

$$v\left(x,y,o\right)=0 \tag{43}$$

$$c\left(x,y,o\right)=0.5\left(1+l\left(x,y;2,0.5\right)\right) \tag{44}$$

The horizontal component of velocity is perturbed by a sine wave with the amplitude of 0.01 as suggested by [14]. The concentration between two fluids at the interface is defined by a hyperbolic tangent function with the width of ϵ. The computational domain is $\Omega=[0,1]\,x\,[0,1]$. The periodic boundary conditions are used at the left and right of the domain ($x=0$ and $x=L$), and $\frac{\partial u}{\partial y}=v=\frac{\partial p}{\partial y}=\frac{\partial c}{\partial y}=\frac{\partial \mu}{\partial y}=0$ are applied at the top and bottom of the domain.

3.2. Numerical solution

The fractional step method is used to solve the momentum equations. The implementation of the fractional step method for the modification of Navier–Stokes equations is conducted by using two steps. In the first step, the pressure term is dropped from the momentum equations, and the equations become:

$$\frac{\partial\tilde{u}}{\partial t}+u\frac{\partial\tilde{u}}{\partial x}+v\frac{\partial\tilde{u}}{\partial y}-\frac{\eta}{Re.\rho\left(c\right)}\left(\frac{\partial^2\tilde{u}}{\partial x^2}+\frac{\partial^2\tilde{u}}{\partial y^2}\right)=\frac{\mathrm{SF}\left(\mathrm{c}\right)}{\rho\left(c\right)} \tag{45}$$

$$\frac{\partial\tilde{v}}{\partial t}+u\frac{\partial\tilde{v}}{\partial x}+v\frac{\partial\tilde{v}}{\partial y}-\frac{\eta}{Re.\rho\left(c\right)}\left(\frac{\partial^2\tilde{v}}{\partial x^2}+\frac{\partial^2\tilde{v}}{\partial y^2}\right)=\frac{\mathrm{SF}\left(\mathrm{c}\right)}{\rho\left(c\right)}+\frac{\mathrm{g}}{Fr^2} \tag{46}$$

where $\tilde{u}$ and $\tilde{v}$ are the intermediate velocity components. Next, the implicit Euler scheme is used to solve the momentum equation.

$$\frac{\tilde{u}}{\Delta t}+u\frac{\partial\tilde{u}}{\partial x}+v\frac{\partial\tilde{u}}{\partial y}-\frac{\eta}{Re.\rho\left(c\right)}\left(\frac{\partial^2\tilde{u}}{\partial x^2}+\frac{\partial^2\tilde{u}}{\partial y^2}\right)=\frac{u^n}{\Delta t}+\frac{\mathrm{SF}\left(\mathrm{c}\right)}{\rho\left(c\right)} \tag{47}$$

$$\frac{\tilde{v}}{\Delta t}+u\frac{\partial\tilde{v}}{\partial x}+v\frac{\partial\tilde{v}}{\partial y}-\frac{\eta}{Re.\rho\left(c\right)}\left(\frac{\partial^2\tilde{v}}{\partial x^2}+\frac{\partial^2\tilde{v}}{\partial y^2}\right)=\frac{v^n}{\Delta t}+\frac{\mathrm{SF}\left(\mathrm{c}\right)}{\rho\left(c\right)}+\frac{\mathrm{g}}{Fr^2} \tag{48}$$

Here the calculated intermediate velocities of $\tilde{u}$ *and* $\tilde{v}$ from Eqs. (47) and (48) do not satisfy the continuity equation. For this reason, the intermediate velocity components need to be corrected by using the pressure term from the Poisson equation. These are obtained from the combination of momentum and continuity equations and is expressed as

$$\frac{\partial^2p^{n+1}}{\partial x^2}+\frac{\partial^2p^{n+1}}{\partial y^2}=\frac{\rho(c)}{\Delta t}\left(\frac{\partial\tilde{u}}{\partial x}+\frac{\partial\tilde{v}}{\partial y}\right) \tag{49}$$

If pressure term $\left(p^{n+1}\right)$ is known, the correct velocities $\left(u^{n+1}\ and\ v^{n+1}\right)$ can be determined by using the below equations:

$$u_i^{n+1}=\tilde{u}-\frac{\Delta t}{\rho(c)}\frac{\partial p^{n+1}}{\partial x} \tag{50}$$

$$v_i^{n+1}=\tilde{v}-\frac{\Delta t}{\rho(c)}\frac{\partial p^{n+1}}{\partial y} \tag{51}$$

After completing the calculations of u, v, and p, the Cahn–Hilliard equation is solved independently in order to find the concentration c and chemical potential μ. It is calculated by using the semi implicit scheme, as follows:

$$\frac{c^{n+1}-c^n}{\Delta t}-\frac{1}{Pe}\nabla^2\mu^{n+1}=-\nabla\left(c.u\right)^n \tag{52}$$

$$\mu^{n+1}+\epsilon^2\nabla^2c^{n+1}=f\left(c^n\right)+\beta\left(c^n\right) \tag{53}$$

The concentration c and the chemical potential μ are solved in a coupled way and the following equation is obtained.

$$\begin{bmatrix}\frac{I}{\Delta t} & -\frac{1}{Pe}\nabla^2\\ \epsilon^2\nabla^2 & I\end{bmatrix}\begin{bmatrix}c^{n+1}\\ \mu^{n+1}\end{bmatrix}=\begin{bmatrix}\frac{c^n}{\Delta t}-\nabla\left(c.u\right)^n\\ f\left(c^n\right)+\beta\left(c^n\right)\end{bmatrix} \tag{54}$$

3.3. Discretization

The momentum equation in the x-axis direction is discretized by using a linear combination of RBF and is expressed in the form of the below equations

$$\tilde{u} = \sum_{j=1}^{N} \alpha_{1j}\varphi \tag{55}$$

$$\frac{1}{\Delta t}\sum_{j=1}^{N} \alpha_{1j}\varphi + u^n \sum_{j=1}^{N} \alpha_{1j}\frac{\partial \varphi}{\partial x} + v^n \sum_{j=1}^{N} \alpha_{1j}\frac{\partial \varphi}{\partial y} - \frac{\eta}{Re.\rho\,(c)}\left(\sum_{j=1}^{N} \alpha_{1j}\frac{\partial^2 \varphi}{\partial x^2} + \sum_{j=1}^{N} \alpha_{1j}\frac{\partial^2 \varphi}{\partial y^2}\right) = \frac{u^n}{\Delta t} + \frac{SF(c)}{\rho(c)} \tag{56}$$

where $\tilde{u}$ is the horizontal component of intermediate velocity, α is the expansion coefficient and φ is the basis function.

The momentum equation at the y-axis is discretized also by using a linear combination of RBF and is expressed in the form of the below equations

$$\tilde{v} = \sum_{j=1}^{N} \alpha_{2j}\varphi \tag{57}$$

$$\frac{1}{\Delta t}\sum_{j=1}^{N} \alpha_{2j}\varphi + u^n \sum_{j=1}^{N} \alpha_{2j}\frac{\partial \varphi}{\partial x} + v^n \sum_{j=1}^{N} \alpha_{2j}\frac{\partial \varphi}{\partial y} - \frac{\eta}{Re.\rho\,(c)}\left(\sum_{j=1}^{N} \alpha_{2j}\frac{\partial^2 \varphi}{\partial x^2} + \sum_{j=1}^{N} \alpha_{2j}\frac{\partial^2 \varphi}{\partial y^2}\right) = \frac{v^n}{\Delta t} + \frac{SF\,(c)}{\rho(c)} + \frac{g}{Fr^2} \tag{58}$$

where $\tilde{v}$ is the vertical component of intermediate velocity, α is the expansion coefficient and φ is the basis function. Next, the discretized forms for the pressure equations are defined as:

$$\tilde{u} = \sum_{j=1}^{N} \alpha_{1j}\varphi, \qquad \tilde{v} = \sum_{j=1}^{N} \alpha_{2j}\varphi, \qquad p^{n+1} = \sum_{j=1}^{N} \alpha_{3j}\varphi \tag{59}$$

$$\sum_{j=1}^{N} \alpha_{3j}\frac{\partial^2 \varphi}{\partial x^2} + \sum_{j=1}^{N} \alpha_{3j}\frac{\partial^2 \varphi}{\partial y^2} = \frac{\rho\,(c)}{\Delta t}\left(\sum_{j=1}^{N} \alpha_{1j}\frac{\partial \varphi}{\partial x} + \sum_{j=1}^{N} \alpha_{2j}\frac{\partial \varphi}{\partial y}\right) \tag{60}$$

Then, the discretized forms for the velocity correction equations are defined as:

$$u_i^{n+1} = \tilde{u} - \Delta\frac{t}{\rho(c)}\sum_{j=1}^{N} \alpha_{3j}\frac{\partial \varphi}{\partial x} \tag{61}$$

$$v_i^{n+1} = \tilde{v} - \frac{\Delta t}{\rho(c)}\sum_{j=1}^{N} \alpha_{3j}\frac{\partial \varphi}{\partial y} \tag{62}$$

The Cahn–Hilliard equations are also discretized using a linear combination of RBFs and are expressed as follows:

$$c^{n+1} = \sum_{j=1}^{N} \alpha_{4j}\varphi, \quad \mu^{n+1} = \sum_{j=1}^{N} \alpha_{5j}\varphi \tag{63}$$

$$\frac{1}{\Delta t}\sum_{j=1}^{N} \alpha_{4j}\varphi - \frac{1}{Pe}\left(\sum_{j=1}^{N} \alpha_{5j}\frac{\partial^2 \varphi}{\partial x^2} + \sum_{j=1}^{N} \alpha_{5j}\frac{\partial^2 \varphi}{\partial y^2}\right) = \frac{c^n}{\Delta t} - u^n \sum_{j=1}^{N} \alpha_{4j}\frac{\partial \varphi}{\partial x} - v^n \sum_{j=1}^{N} \alpha_{4j}\frac{\partial \varphi}{\partial y} \tag{64}$$

$$\sum_{j=1}^{N} \alpha_{5j}\varphi + \epsilon^2\left(\sum_{j=1}^{N} \alpha_{4j}\frac{\partial^2 \varphi}{\partial x^2} + \sum_{j=1}^{N} \alpha_{4j}\frac{\partial^2 \varphi}{\partial y^2}\right) = f\left(c^n\right) + \beta\left(c^n\right) \tag{65}$$

where Δt denotes the time step, superscript $n+1$ is the unknown value to be solved, and superscript n is the current known value.

4. Results and discussions

In the next section, the meshless method on the basis of the RBF in the framework of the fractional step method is used to solve the KHI of two incompressible and immiscible fluids. The instability is described by the modification of 2-D

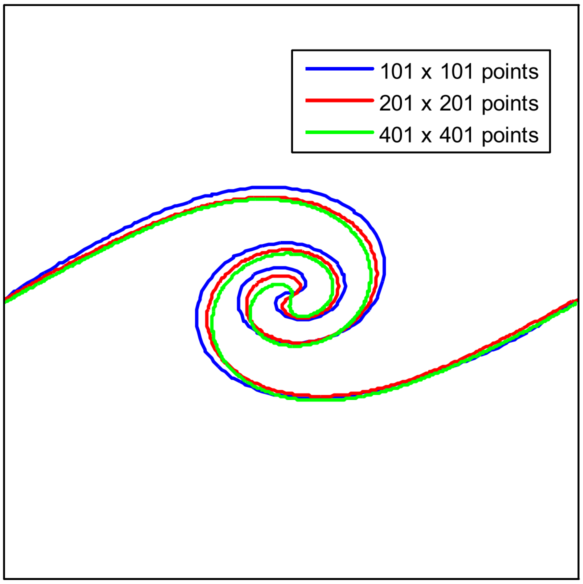

Fig. 3. Point independence test: interface evolutions using 101 × 101 (blue line), 201 × 201 (red line), and 401 × 401 (green line) point distribution at dimensionless time $t = 1$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

incompressible Navier–Stokes equations as shown in Eqs. (31)–(32). Here the Cahn–Hilliard equation is used for capturing the interface of working fluids.

In the present numerical study, the point independence tests were carried-out firstly in order to determine the point independent solution, whereas the numerical results will not change when the number of points are increased. The numerical simulations were carried out under the same of initial conditions (Eqs. (42)–(44)) and numerical properties such as density, viscosity, *Re*, time step, etc. Three selected point distributions (101 × 101, 201 × 201 and 401 × 401) were used in the point independence test. The domain was decomposed into 10 × 10 subdomains for point distribution of 101 × 101 and 15 × 15 subdomains for point distribution of 201 × 201 and 401 × 401. The numerical parameters are $\Delta t = 0.0005$, $\epsilon = 0.01/\sqrt{2}$, $\mathrm{Re} = 5000$, $\mathrm{Fr} = 1$, $\rho_1 : \rho_2 = 0.99 : 1$, $\eta_1 : \eta_2 = 1 : 1$ and $Pe = 10/\epsilon$. To ease the understanding of the reader the pseudo codes of the KHI are also given in the appendix of the manuscript. The results are shown in Fig. 3 for the case of interface evolutions. From the figure, it is shown that only a small difference on the interface evolution exists in the numerical results when the point distributions were 201 × 201 and 401 × 401. This means that the point distribution of 201 × 201 that is decomposed into 15 × 15 subdomains is an optimum point distribution for the simulations.

In order to illustrate the accuracy of the RBF method, the mixing of two immiscible fluids in the rectangular cavity is considered as a test case. The interface evolutions during the mixing of two immiscible fluids of the present study are compared with the results of Khatavkar [36] under the same domain and boundary conditions. The numerical parameters were $Pe = 10^5$, $\mathrm{Ch} = 0.01$, the viscosity ratio $\lambda = 1$, and the Capillary number $Ca = 500$. The initial conditions of the concentration and velocity were obtained by the following equations

$$c = tanh\left(\frac{y - \frac{H}{2}}{\sqrt{2}Ch}\right) \tag{66}$$

$$u = 16\left(\frac{x}{L}\right)^2\left(1 - \frac{x}{L}\right)^2, \;\; v = 0, \;\; \text{at } y = 0 \tag{67}$$

where $L = 1$ and $H = 1$.

The comparisons of the interface evolution ranging from $t = 5$ to $t = 15$ are shown in Fig. 4. The figures reveals that the interface evolution obtained from the present works is in good agreement with that of [36]. This means that the implemented RBF method has a good ability to predict the interface evolution fields during the mixing of two immiscible fluids in a rectangular cavity.

4.1. Linear growth rate of the Kelvin–Helmholtz instability

In the next section, the linear growth rate of KHI as a function of density ratio is studied in order to investigate the accuracy of the present numerical method. The two-fluid system of KHI is perturbed by applying a sinusoidal disturbance

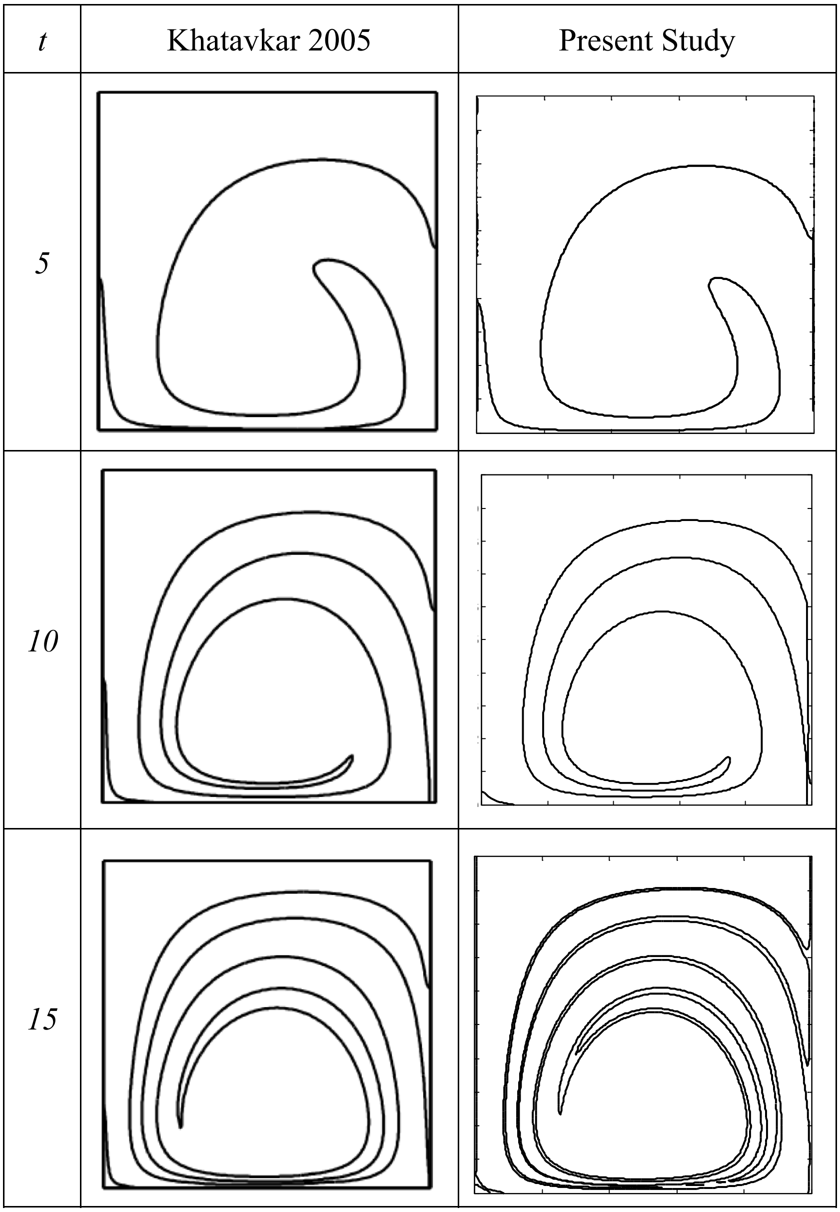

Fig. 4. The comparison of interface evolution of mixing two fluids between the present study and Khatavkar [36] for $Pe = 10^5$, $Ch = 0.01$, $\lambda = 1$, and $\mathrm{Ca} = 500$.

on the interface of working fluids in the form of:

$$\zeta = \zeta_o e^{i(kx-\omega t)} \tag{68}$$

where ζ is the wave amplitude of the interface, which is a function of horizontal direction x, time t, angular frequency ω, the wave number k, and the amplitude of the initial sinusoidal disturbance ζ_o. In the case of both gravitational and surface tension forces are present, the eigenvalue condition [17] is as follows:

$$\omega = k\frac{\rho_1 U_{1+}\rho_2 U_2}{\rho_1+\rho_2} \pm i\sqrt{\frac{k^2\rho_1\rho_2\left(U_1-U_2\right)^2}{\left(\rho_1+\rho_2\right)^2} - \frac{gk\left(\rho_1-\rho_2\right)}{\left(\rho_1+\rho_2\right)^2} - \frac{\sigma k^3}{\rho_1+\rho_2}} \tag{69}$$

The imaginary parts of ω have the linear growth rate $G_e = \mathrm{Im}(\omega)$. In the case of $g = 0$ and $\sigma = 0$, the formulation of the analytical growth rate (G_e) is defined as follows:

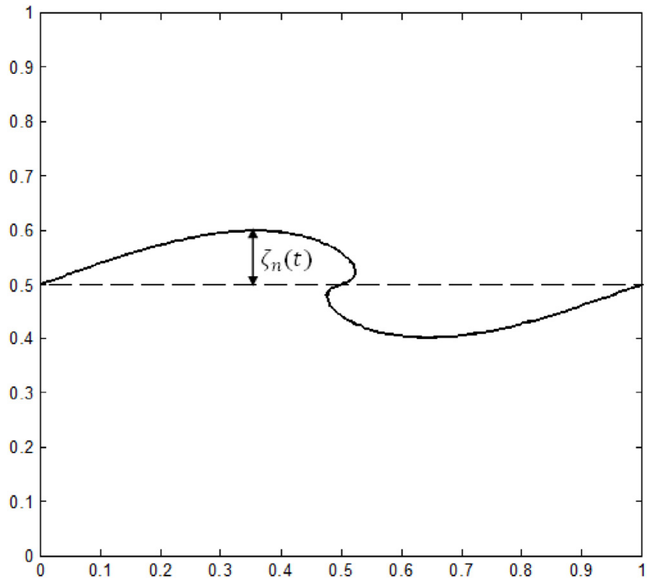

Fig. 5. The amplitude of the interface $\zeta_n\,(t)$ at dimensionless time t.

$$G_e = \mathrm{Im}\,(\omega) = \frac{4\pi\sqrt{r}}{1+r}, \qquad r = \frac{\rho_1}{\rho_2} \tag{70}$$

where r is the density ratio.

In order to calculate the numerical growth as a function of the density ratio, the initial conditions are considered by Eqs. (42) to (44). The numerical parameters are $\Delta t = 0.00025$, $\epsilon = 0.02/\sqrt{2}$, $\mathrm{Re} = 5000$, $\mathrm{g} = 0$, $\mathrm{Fr} = 1$, $Pe = 10/\epsilon$, and the point distributions are 101×101, 201×201 and 401×401. The domain was decomposed into 15×15 subdomains. For numerical investigation, the numerical growth rate G_n is as follows [17]:

$$G_n = \frac{(\zeta_n\,(t)\,/\zeta_o) - 1}{t} \tag{71}$$

where $\zeta_n\,(t)$ is the amplitude of the interface at dimensionless time t, and ζ_o is the initial amplitude of the applied disturbance ($\zeta_o = 0.01$).

Fig. 5 illustrates the amplitude of the interface $\zeta_n\,(t)$ used in the present numerical study. Here, the dimensionless time t is taken when the wave amplitude reaches up about 10% of the domain height ($\zeta_n(t) \cong 0.1H$). The comparison between the obtained result from the present study for the numerical resolution of 101×101 (red asterisk), 201×201 (blue circle) and 401×401 (green diamond) and the analytical growth rate is shown in Fig. 6. The average errors are 7.32%, 5.42%, and 6.31% for the point distribution of 101×101, 201×201, and 401×401, respectively.

By a close investigation of Fig. 6, it can be seen that the results are not converging toward the analytical results at higher density ratios, regarding the increasing of the points. In RBF method, when the points increase, the shape parameter must be adjusted to avoid ill-conditioned of the matrix. If the shape parameter adjusted too much, the accuracy of the RBF method will decrease [16]. Therefore, it is noticed that the 201×201 point distribution is an optimum point distribution for the simulation.

4.2. Kelvin–Helmholtz instability of two-component fluids

Point distribution for solving the KHI of two-component fluids was conducted by 201×201 point numbers arranged in a uniform grid, in which the point separation distance was $L/200$. In this case, L and H were unity. From the point independence test conducted above, the 201×201 point numbers were the optimum point distribution for the calculations. The domain was decomposed into 15×15 subdomains. The two-fluid system has a viscosity ratio of $\eta_1{:}\,\eta_2 = 1{:}\,1$ and density ratio of $\rho_1{:}\,\rho_2 = 0.99{:}\,1$. The other parameters were $\Delta t = 0.0005$, $\epsilon = 0.01/\sqrt{2}$, $\mathrm{Re} = 5000$, $\mathrm{Fr} = 1$ and $Pe = 10/\epsilon$.

In the present study, the initial conditions of the velocity and concentration were obtained from Eqs. (42) to (44). The periodic boundary conditions were implemented at $x = 0$, and $x = L$. The $\frac{\partial u}{\partial y} = v = \frac{\partial p}{\partial y} = \frac{\partial c}{\partial y} = \frac{\partial \mu}{\partial y} = 0$ were applied at the top and bottom of the domain. In order to ensure the implementation of the periodic boundary conditions, the ghost node strategy was used. It enables randomly placed points near the boundary and also satisfies the boundary conditions accurately as suggested by Chinchapatnam [25].

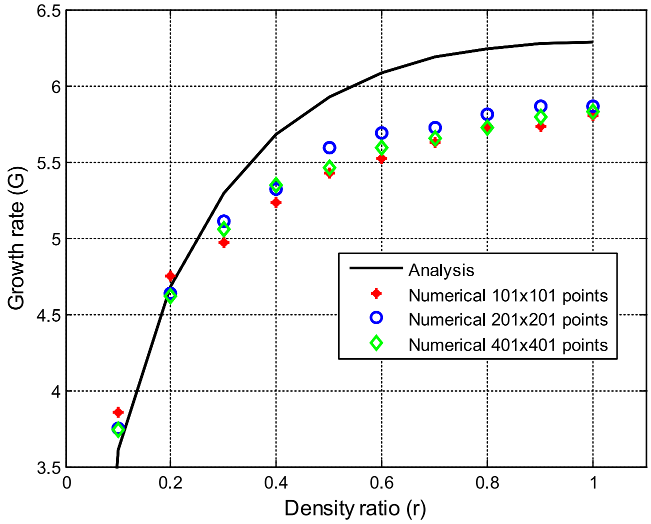

Fig. 6. The numerical growth rate obtained from the present numerical study for the point distributions of 101 × 101, 201 × 201, 401 × 401, and analytical linear growth rate of KHI.

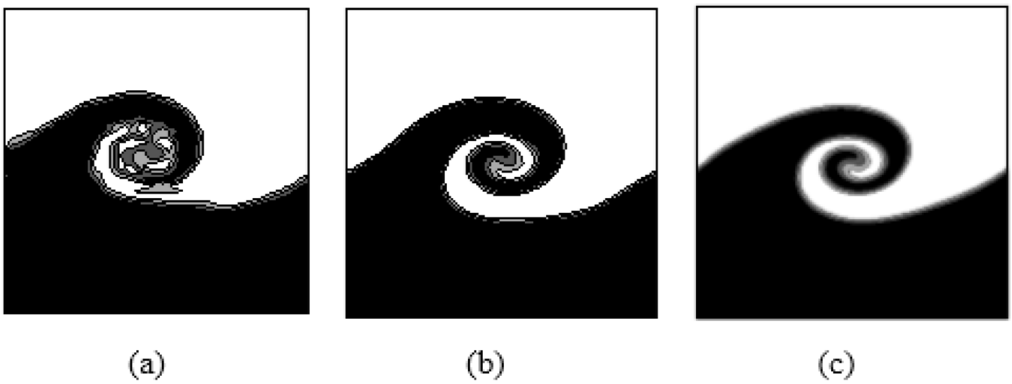

Fig. 7. The interface evolution for different numerical resolutions, (a) 51 × 51, (b) 81 × 81, (c) 101 × 101, at $t = 1$.

The resolution of RBF method refers to the number of point distributions in the computational domain. In order to study the minimum numerical resolution required to activate the instability, the KHI of two-component was simulated with the numerical resolution of 51 × 51, 81 × 81, and 101 × 101. The numerical results at $t = 1$ are shown in Fig. 7. The results indicate that the interface evolution becomes unstable for the numerical resolution of 51 × 51. For a higher numerical resolution of 81 × 81, the roll up of interface evolution becomes smoother. The most stable interface evolution can be reached for the numerical resolution of 101 × 101. It can be concluded that in the range of the tested point distributions in the present study, the minimum resolution of radial basis function method to produce the stable interface evolution of KHI is of 81 × 81 point distribution.

Fig. 8 shows the calculation results of the interface evolution as a function of dimensionless time in the two-component fluids of KHI obtained from the present study with numerical resolution of 101 × 101, 201 × 201, and 401 × 401. As shown in the figure, the interface is flat at $t = 0.0$ and parallel to the direction of velocity shear. At $t = 0.6$–0.7, the interfaces start to wrinkle due to the initial perturbation and velocity shear. A rolled-up vortex exists around the initial interface at $t = 1.0$. Then, a large vortex is observed and a mixing layer is formed around the interface *at* $t = 1.2$–1.4.

The qualitative behavior of the interface evolution for the low (101 × 101), medium (201 × 201), and high (401 × 401) numerical resolutions is also considered herein. The evolutions appear to be similar when the numerical resolution is changed. Close investigation on the figure, at $t = 1.4$, the interface evolution produces more roll up at the low numerical solution. In addition, it can be seen that the interface evolution for medium numerical resolution (201 × 201) has a good agreement with the previous results from Lee et al. [14].

Next, the effects of the interface thickness (ϵ) on the interface evolution are considered. Fig. 9 shows the numerical result of the effect of ϵ on the interface evolutions at dimensionless time $t = 1$. In the figure, (a), (b), and (c) correspond to the cases of $\epsilon = 0.02/\sqrt{2}$, $\epsilon = 0.01/\sqrt{2}$, and $\epsilon = 0.005/\sqrt{2}$, respectively. Close investigation on the figure reveals that

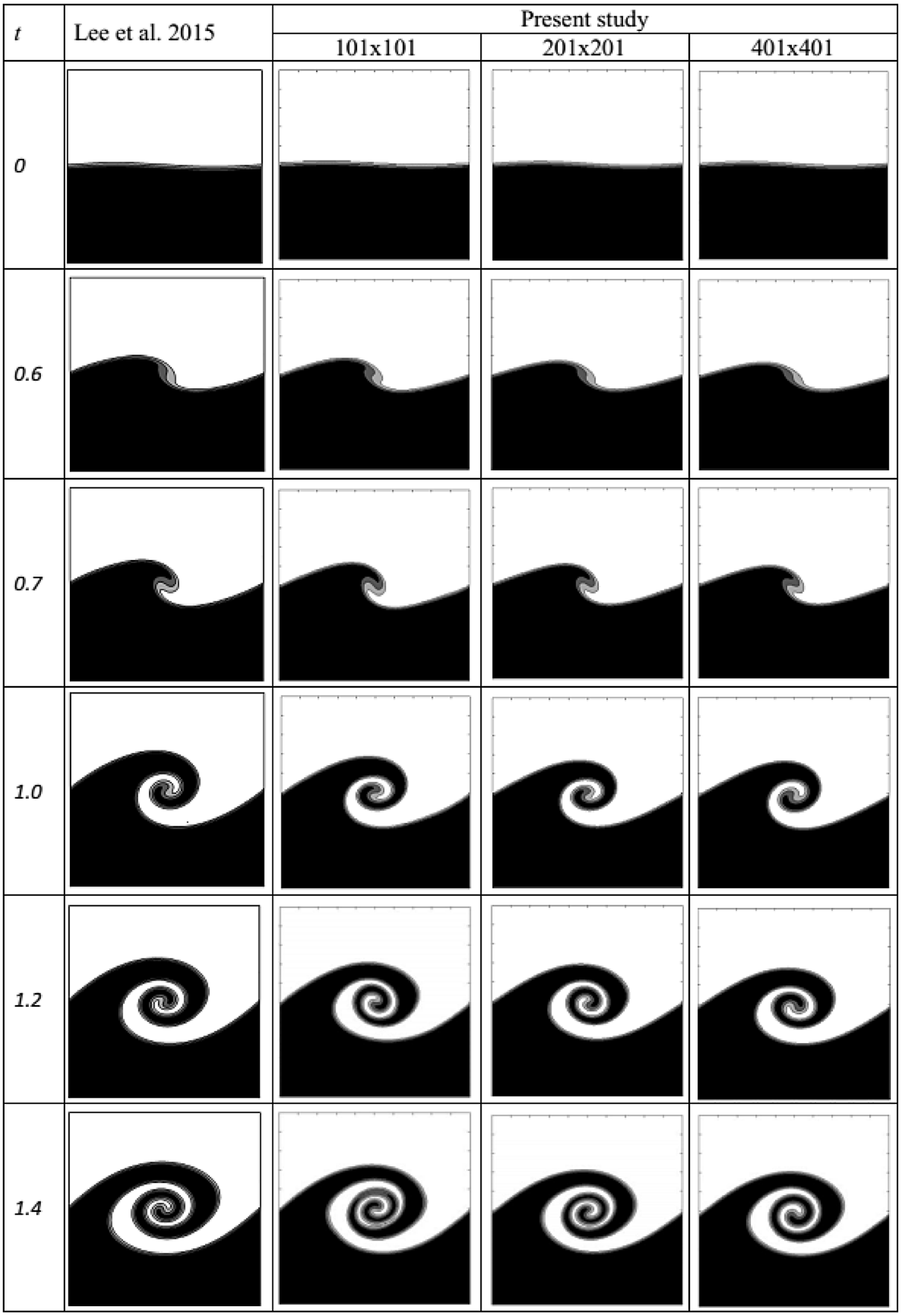

Fig. 8. The comparison of interface evolution between the obtained results from present study for point distribution of 101 × 101, 201 × 201, and 401 × 401 and the numerical results from Lee et al. [14] under the same flow conditions of $\Delta t = 0.0005$, $\epsilon = 0.01/\sqrt{2}$, $Re = 5000$, $Fr = 1$ and $Pe = 10/\epsilon$.

the interface evolutions are affected by ϵ. The decrease of the ϵ produces non-smooth concentration profile, whereas the increase of the ϵ produces too much diffusion. Otherwise, if the $\epsilon = 0.01/\sqrt{2}$ is taken, it produces a smooth interfacial

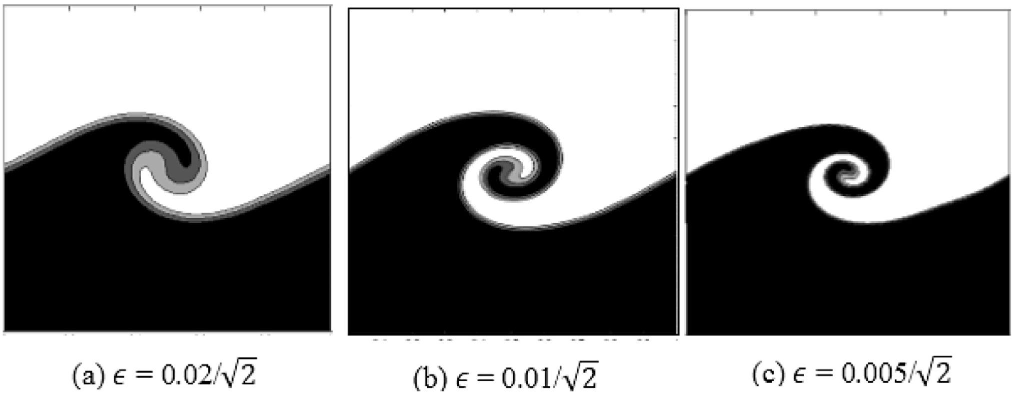

Fig. 9. The effect of interface thickness ϵ on the interface evolution at dimensionless time of $t = 1$.

transition, and the rolling interface profile follows the flow field. This means that the interface evolutions are sensitive to the change of ϵ.

4.3. Kelvin–Helmholtz instability of three-component fluids

In the next section, the KHI at the interface among three components of horizontal parallel fluids under the difference of the velocities and densities is investigated by using the present method. The hyperbolic tangent function was used to determine the velocities and concentration of the initial conditions. The initial sinusoidal perturbations were applied to the working fluids interface. Point distribution for solving KHI of three-component fluids was conducted by 201 × 201 point numbers arranged in a uniform grid, where the point separation distance was $L/200$. The domain was decomposed into 15 × 15 subdomains. In this case, L and H were unity. To investigate the effect of density ratio on the interface evolution, three variations of density ratio were implemented. The variations of the density ratio were $\rho_1\colon \rho_2\colon \rho_3 = 0.98\colon 0.99\colon 1$, $\rho_1\colon \rho_2\colon \rho_3 = 0.4\colon 0.7\colon 1$, and $\rho_1\colon \rho_2\colon \rho_3 = 0.1\colon 0.55\colon 1$. The system has a viscosity ratio of $\eta_1\colon \eta_2\colon \eta_3 = 1\colon 1\colon 1$. The other parameters were $\Delta t = 0.001$, $\epsilon = 0.006/\sqrt{2}$, Re $= 5000$, Fr $= 1$ and $Pe = 10/\epsilon$.

The initial conditions of the velocity and concentration were obtained by the following equations.

$$c_1 (x, y, o) = 0.5 (1 + l (x, y; 4, 2/3)) \tag{72}$$

$$c_2 (x, y, o) = 0.5 (1 + l (x, y; 4, 1/3)) - c_1 (x, y, o) \tag{73}$$

$$u (x, y, o) = 1 + l (x, y; 4, 2/3) - l (x, y; 4, 1/3) \tag{74}$$

$$v (x, y, o) = 0 \tag{75}$$

The transition layers between two fluids at each interface are defined by a hyperbolic tangent profile (Eqs. (72) and (73)). Eqs. (74) and (75) define the velocities of three components of horizontal parallel fluids. The amplitude of sinusoidal perturbation was set to 0.01.

Fig. 10 illustrates the effect of density ratio on the interface evolution. In the figures, (a), (b), and (c) correspond to the cases of $\rho_1\colon \rho_2\colon \rho_3 = 0.1\colon 0.55\colon 1$, $\rho_1\colon \rho_2\colon \rho_3 = 0.4\colon 0.7\colon 1$, and $\rho_1\colon \rho_2\colon \rho_3 = 0.98\colon 0.99\colon 1$ respectively. The figure indicates that the decrease of the density ratio, the transition from linear to non-linear stage is expedited to the later simulation time.

In order to investigate the influence of magnitude velocity difference on the interface evolution, the same parameters and the initial conditions of the concentration (Eqs. (72) and (73)) were used. Three initial conditions of the horizontal velocities are obtained from the following equations.

$$u (x, y, o) = 1 + 0.5 (l (x, y; 4, 2/3) - l (x, y; 4, 1/3)) \tag{76}$$

$$u (x, y, o) = 1 + (l (x, y; 4, 2/3) - l (x, y; 4, 1/3)) \tag{77}$$

$$u (x, y, o) = 1 + 1.5 (l (x, y; 4, 2/3) - l (x, y; 4, 1/3)) \tag{78}$$

$$v (x, y, o) = 0 \tag{79}$$

A sinusoidal perturbation with an amplitude of 0.01 was also applied at the initial velocities (Eqs. (76) to (78)) to generate the roll-up at the interface. The results are shown clearly in Fig. 11. In the figure, (a), (b), and (c) correspond to the horizontal velocities which are defined by Eqs. (76), (77), and (78) respectively. From the figure, it is noticed that the roll-up at the interface increases with the increase of the horizontal velocity difference.

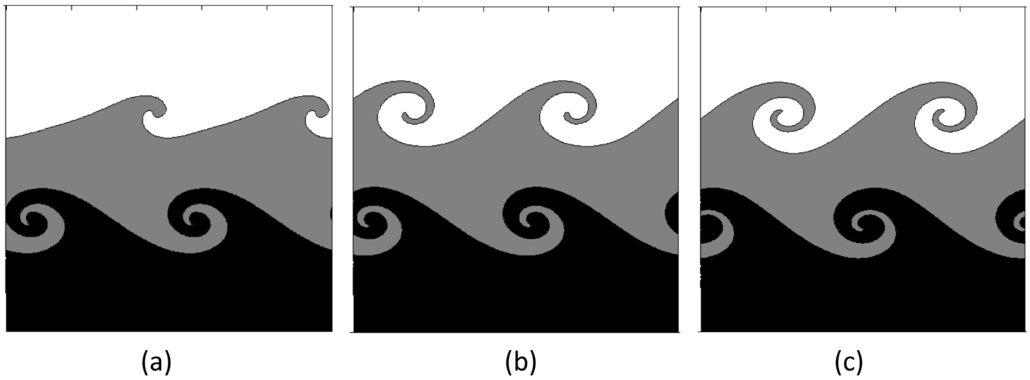

Fig. 10. The effect of the density ratio on the interface evolution at dimensionless time of t = 0.5, (*a*) $\rho_1 : \rho_2 : \rho_3 = 0.1 : 0.55 : 1$, (*b*) $\rho_1 : \rho_2 : \rho_3 = 0.4 : 0.7 : 1$ and (*c*) $\rho_1 : \rho_2 : \rho_3 = 0.98 : 0.99 : 1$.

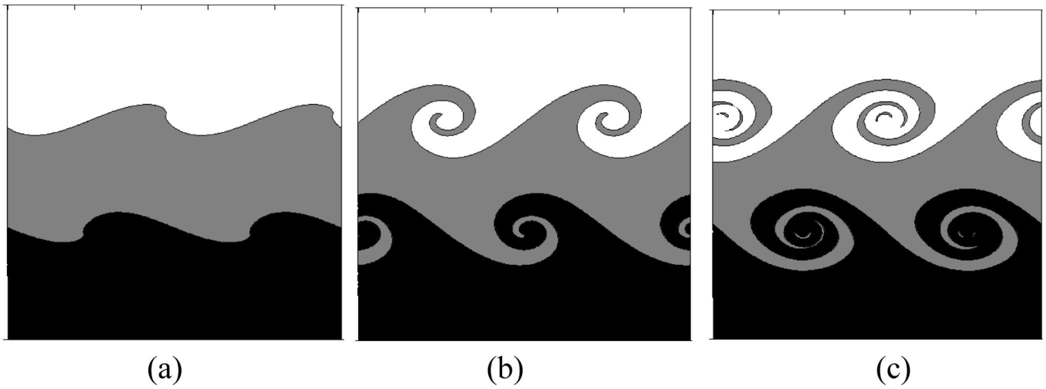

Fig. 11. The effect of the magnitude velocity difference on the interface evolution at dimensionless time of $t = 0.5$ for the initial horizontal velocities in (a) Eq. (76), (b) Eq. (77), and (c) Eq. (78).

4.4. Kelvin–Helmholtz instability on the complex domains

In the present study, the geometry, boundary conditions and point distribution for the complex domains were also considered as shown in Fig. 12. For the complex domain, the walls are subjected to zero velocity boundary conditions. The pressure (p), concentration (c) and chemical potential (μ) are defined as Neumann boundary conditions $\left(\frac{\partial p}{\partial n} = \frac{\partial c}{\partial n} = \frac{\partial \mu}{\partial n} = 0\right)$, where n is normal vector to the boundary. Fig. 13 shows the simulation results of the interface evolution ranging from $t = 0$ to $t = 0.6$ for KHI on the circle, ellipse and trapezoidal domains to represent the interface evolution of the complex domains. By a close inspection of the figure, it is noticed that the RBF method is a reliable method to solve KHI on the complex domains. The classical methods, such as finite difference method (FDM) may produce the troublesome when is applied to simulate the KHI on the complex domains with irregular boundaries.

5. Conclusions

The interface evolution of the KHI in a rectangular cavity and complex domains was investigated numerically by using RBF and domain decomposition method. The phenomena around KHI were modeled by the modification of Navier–Stokes equations. Here the convective Cahn–Hilliard equations were applied to capture the interface between the fluids. The numerical method on the basis of the implicit Euler scheme for temporal discretization and RBF method for spatial discretization were also utilized. The fractional step method was employed to solve the modification of the Navier–Stokes equations. The velocity fields obtained from the present works are in good agreement with the benchmark test results. Next, the interface evolutions obtained from the present numerical study agree well with those from the finite difference method.

The effects of the interface thickness, density ratio and magnitude of velocity difference on the KHI were also investigated numerically. The decrease of the interface thickness produces non-smooth concentration profile and the increase of the interface thickness produces too much diffusion. It was shown also that the increase of the density ratio reduces the growth of the KHI. And finally, the interface rolls-up is strongly affected by the magnitude of initial horizontal velocity difference.

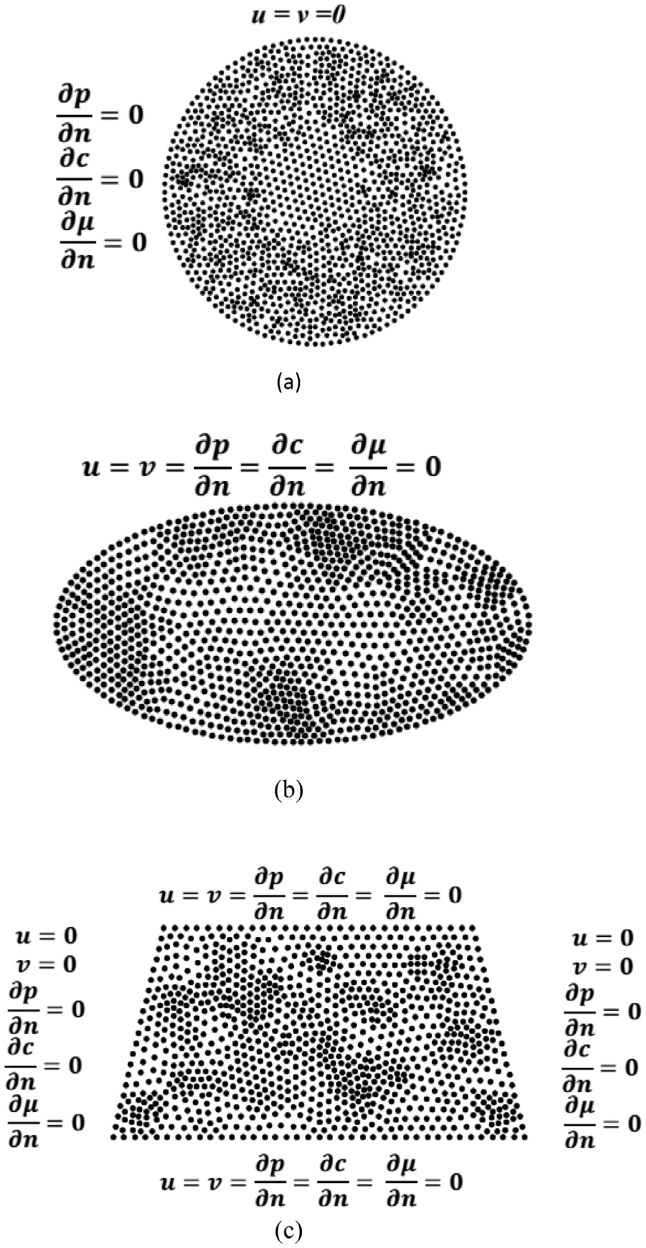

Fig. 12. The geometry, boundary conditions, and point distribution on complex computational domains.

The numerical results showed the progress in the simulation of KHI by using RBF method. The present study shows also that the RBF method is a reliable method to solve KHI on the complex domains. The numerical method will also enable the solving of many problems relating to the two-phase flow, such as Rayleigh–Taylor instability, droplet evaporation, Taylor bubble etc. The method requires a large computational time when calculating large matrix coefficient. Furthermore, additional improvement of the numerical method should be carried out in the near future to reduce the computational time. Therefore, it is necessary to use graphics processing units for parallel computing in order to reduce the computational time.

CRediT authorship contribution statement

Eko Prasetya Budiana: Conceptualization, Methodology, Investigation, Programming, Writing - original draft, Formal analysis. **Pranowo:** Investigation, Data checking, Formal analysis. **Indarto:** Resources, Writing - review & editing, Funding acquisition, Supervision. **Deendarlianto:** Analysis validation, Writing - review & editing, Resources, Funding acquisition, Supervision.

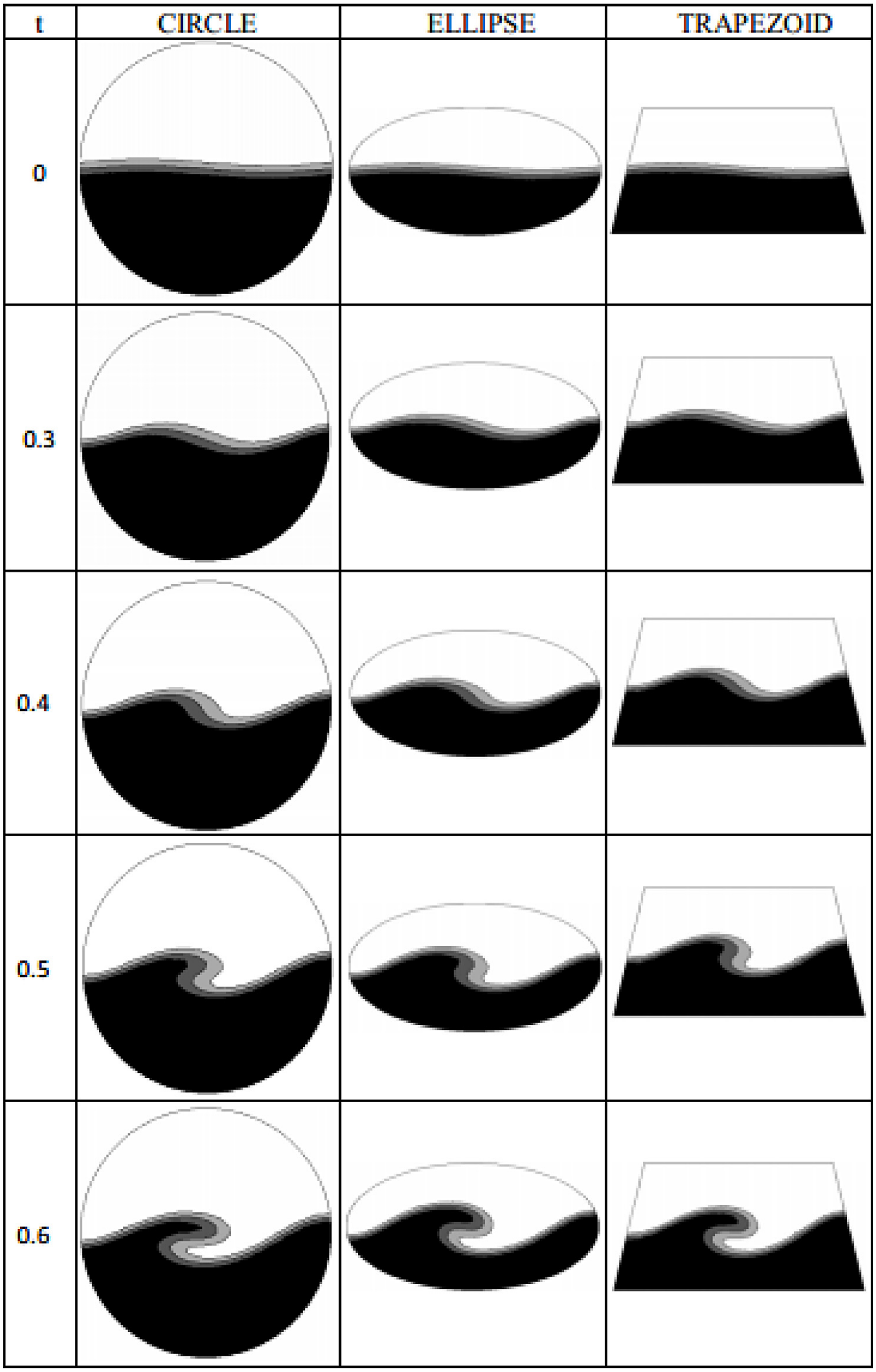

Fig. 13. The interfacial behaviors during KHI on the complex computational domains.

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Appendix

PSEUDO CODE OF KHI

```
MAIN PROGRAM
# Set Function
fC(x) = x*(x-0.5)*(x-1)

# INPUT
# Coordinates, number of nodes and number of sub domains
L, H                 (Length and width of domain)
Nx, Ny               (Number of nodes in x and y direction)
Ng                   (Number of ghost nodes)
Np = (Nx+2*Ng)*Ny    (Number of total nodes)
ndX, ndY             (Number of sub domains in x and y direction)
dX = L/(Nx-1)        (Distance between two nodes in x direction)
dY = H/(Ny-1)        (Distance between two nodes in y direction)
x,y, sp              (Node coordinates and shape parameter)
dt, tmax             (Time step and final time)


Re, Pe, Ch, Fr,g     (Non dimensional numbers &gravitational
acceleration)
ro1, nu1             (density and viscosity of lighter fluid)
ro2, nu2             (density and viscosity of heavier fluid)

# SET COORDINATES OF THE DOMAIN

for i = 1 to (Nx+2*Ng)
      for j = 1 to Ny
                k = (i-1)*Mi+j
          xx(i,j) = (0-Ng*dX)+(j-1)*dX
          yy(i,j) = (i-1)*dY
             x(k) = xx(i,j)
             y(k) = yy(i,j)
      end
end

# SET INITIAL CONDITION
# Refer to Eqs. 60-62
for i =1 to Np
 c(i) = 0.5*(1+tanh((y(i)-H/2-
0.01*sin(pi()*x(i)/(L/nw)))/(2*sqrt(2)*Ch)))

 U(i) = tanh((y(i)-H/2-0.01*sin(pi()*x(i)/(L/nw)))/(2*sqrt(2)*Ch))
 V(i) = 0
 P(i) = 0
end

# SET SUBDOMAIN AND DIFFERENTIATION MATRIX
# Set left, right, top and bottom margins of each subdomain

 CALL (XL,XR,YB,YT)= MARGINS(x,y,ndX,ndY,dX,dY,L,H,Ng)

for  j = 1 to ndY
  for k = 1to ndX
        i = (j-1)*ndX+k
# Set the coordinates of each subdomain and BC
CALL (A1,A2,A3,A4,AI,n1,n2,n3,n4,ni,nI) =
DDM(x,y,XL,XR,YB,YT,k,j,ndX,ndY,Np)
# Set the differentiation matrix Dx, Dy, Dxx, Dyy of each subdomain
CALL (Dx,Dy,Dxx,Dyy,DD) = RBF_Derivatives_LU(nI,x(AI),y(AI),sp,i)

  end
end
```

```
# ITERATIVE SOLUTION
   t = 0
while (t<tmax)
for j = 1 to ndY
for k = 1 to ndX
      i = (j-1)*ndX+k

# Transfer numerical variables from main domain to subdomain
  V   to  Vs
  U   to  Us
  P   to  Ps
  c   to  cs
 mu   to  mus

   f1 = fC(cs)+fC(1-cs)
   f2 = fC(cs)
   NJ       (number of nodes at each subdomain)
for m = 1 to NJ
  Beta(m)  = -cs(m)*f1(m)               (Refer to eq. 54)
   Rho(m)  = cs(m)*ro1+(1-cs(m))*ro2    (Refer to eq. 55)
     Nu(m) = cs(m)*nu1+(1-cs(m))*nu2    (Refer to eq. 56)
    Nro(m) = Nu(m)/Rho(m)/Re
  end

 CALL (IM) = IDENTITY(NJ)  (set identity matrix IM(NJxNJ))
CALL (uDx) = DOT_ROW(Us,Dx,NJ)      (to calculate uDx)
CALL (vDy) = DOT_ROW(Vs,Dy,NJ)      (to calculate vDy)

CALL  (Dif) = DOT_ROW(Nro,DD,NJ)          (to calculate vDy)
# Set coefficient matrix of momentum eqs. refer to eq. 65
for m = 1 to  NJ
   for m = 1 to  NJ
       DDUV(m,m) = IM(m,m)/dt+uDx(m,m)+vDy(m,m)-Diff(m,m)
   end
end

 # Set right hand side (RHS) of momentum eqs.
 for m = 1 to NJ
   RHSU(m) = Us(m)/dt          (refer to eq. 65)
   RHSV(m) = Vs(m)+g/Fr        (refer to eq. 66)
 end

 # Set left, right, top and bottom BC of momentum eqs.
 Set Dirichlet BC of Us, Vs
 If k = 1   then   (Set periodic BC of Us, Vs at left domain)
 If k = ndX then   (Set periodic BC of Us, Vs at right domain)
 If j = 1   then   (Set Neumann BC of Us at bottom domain)
 If j = ndY then   (Set Neumann BC of Us at top domain)

 # Solve momentum eqs. using LU Decomposition method
 CALL (uu) = LU_SOLVE(DDUV,RHSU,NJ)  (calculate intermediate
 velocity uu)
 CALL (vv) = LU_SOLVE(DDUV,RHSV,NJ)  (calculate intermediate
 velocity vv)

 CALL (Dxu) = CROSS(Dx,uu,NJ,NJ,1)  (to calculate Dxu)
 CALL (Dyv) = CROSS(Dy,vv,NJ,NJ,1)  (to calculate Dyv)

 for m = 1 to NJ
   RHSP(m) = (Dxu(m)+Dyv(m))/dt      (RHS of pressure term, refer
 to eq. 67)
 End
```

```
# Set left, right, top and bottom BC of pressure eqs.
Set Dirichlet BC of Ps
If k = 1   then   (Set periodic BC of Ps at left domain)
If k = ndX then   (Set periodic BC of Ps at right domain)
If j = 1   then   (Set Neumann BC of Ps at bottom domain)
If j = ndY then   (Set Neumann BC of Ps at top domain)

# Set coefficient matrix of pressure eqs. refer to eq. 67
for m = 1 to NJ
 for m = 1 to NJ
   DDP(m,m) = Dxx(m,m)+Dyy(m,m)
 end
end

# Solve pressure eqs. using LU Decomposition method
  CALL (Ps) = LU_SOLVE(DDP,Rhs,NJ)

 CALL (dxp) = CROSS(Dx,Ps,NJ,NJ,1)
 CALL (dyp) = CROSS(Dy,Ps,NJ,NJ,1)
#==================================================================
==
# Pressure correction
  for s1 = 1 to NJ
      Us(s1) = uu(s1)-dt*dxp(s1)          refer to eq. 68
      Vs(s1) = vv(s1)-dt*dyp(s1)          refer to eq. 69
  end

#==================================================================
==
# Set RHS of Cahn-Hilliard eqs.
      f2 = fC(cs(J))
      Dc = CROSS(DD,cs,NJ,NJ,1)

     Dmu = CROSS(DD,mus,NJ,NJ,1)
     Dxc = CROSS(Dx,cs,NJ,NJ,1)
    uDxc = DOT(Us,Dxc,NJ)
     Dyc = CROSS(Dy,cs,NJ,NJ,1)
    vDyc = DOT(Vs,Dyc,NJ)

      for s1 = 1 to NJ
            Rhsc(s1) = cs(s1)/dt-uDxc(s1)-vDyc(s1)    (refer to
eq. 72)
         Rhsc(s1+NJ) =  f2(s1)+Beta(s1)               (refer to
eq. 72)
      end

 # Set Dirichlet BC of cs and mus
 If k = 1   then   (Set periodic BC of cs, mus at left domain)
 If k = ndX then   (Set periodic BC of cs, mus at right domain)
 If j = 1   then   (Set Neumann BC of cs, mus at bottom domain)
 If j = ndY then   (Set Neumann BC of cs, mus at top domain)

#==================================================================
===
for s1 = 1 to NJ                           (refer to eq. 72)
   AA1(s1,J) = IM(s1,J)/dt
   AA2(s1,J) = -DD(s1,J)/Pe
   AA3(s1,J) = Ch^2*DD(s1,J)
   AA4(s1,J) = IM(s1,J)
end

for s1 = 1 to NJ
        AA(s1,J) = AA1(s1,J);
     AA(s1,J+NJ) = AA2(s1,J);
     AA(s1+NJ,J) = AA3(s1,J);
  AA(s1+NJ,J+NJ) = AA4(s1,J);
end
#==================================================================
===
```

```
      Rhsc = LU_SOLVE(AA,Rhsc,2*NJ);
       cs  = Rhsc(1:NJ);
       mus = Rhsc(NJ+1:2*NJ);
#=================================================================
===
# Transfer numerical variables from subdomain to main domain
  Vs   to  V
  Us   to  U
  Ps   to  P
  cs   to  c
 mus   to  mu
#=================================================================
===
    end
    end

   t = t + dt
   endwhile
```

1. SUB PROGRAM **MARGINS**

```
(XL,XR,YB,YT) = MARGINS(x,y,ndX,ndY,dX,dY,L,H,Ng)
     nis1 = 1
     nis2 = 0
       dL = (L+2*Ng*dX)/ndX
       dH = H/ndY
for i = 1 to ndX-1
    XX(i) = int(i*dL/dX)*dX
  XL(i+1) = XX(i)-nis1*dX-dX/2
    XR(i) = XX(i)+nis2*dX+dX/2
end
    XL(1) = (0-Ng*dX)-dX/2
  XR(ndX) = (L+Ng*dX)+dX/2

for i = 1 to ndY-1
    YY(i) = int(i*dH/dY)*dY
  YB(i+1) = YY(i)-nis1*dY-dY/2
    YT(i) = YY(i)+nis2*dY+dY/2
end
    YB(1) = 0 - dY/2
  YB(ndY) = H + dY/2
RETURN
```

2. SUB PROGRAM **DDM**

```
(A1,A2,A3,A4,AA,n1,n2,n3,n4,ni,nA)=
DDM(x,y,XL,XR,YB,YT,i,j,ndX,ndY,Np)
#SET THE COORDINATES OF EACH SUBDOMAIN
 for jj = 1 to ndY
     for ii = 1 to ndX
         k1 = (jj-1)*ndX+ii
         id = 0
       for kk = 1 to Np
        if (x(kk)>XL(ii) & x(kk)<XR(ii) & y(kk)>YB(jj) &
y(kk)<YT(jj)) then
                    id = id+1
              SQ(id,k1) = kk
           end
       end
             nS(k1) = id
     end
 end

#=================================================================
======
```

```
                k = (j-1)*ndX+i
            nA(k) = nS(k)

           for is = 1 to nA(k)
            A(is) = SQ(is,k)
           end

             xmax = x(A(1))
             xmin = x(A(1))
             ymax = y(A(1))
             ymin = y(A(1))

     for is = 2 to nA(k)
          if (x(A(is))>xmax) then  xmax = x(A(is))
          if (x(A(is))<xmin) then  xmin = x(A(is))
          if (y(A(is))>ymax) then  ymax = y(A(is))
          if (y(A(is))<ymin) then  ymin = y(A(is))
     end
#==================================================================
======

# SET THE COORDINATES OF THE BC OF EACH SUBDOMAIN
 if (j>1 & j<ndY) then
        id1 = 0
        id2 = 0
        id3 = 0
        id4 = 0
     for ia = 1 to nA(k)
# THE COORDINATES OF LEFT BC OF SUBDOMAIN
         if (x(A(ia))= xmin) then
                 id1 = id1+1
           A1(id1,1) = A(ia)
         end

# THE COORDINATES OF THE RIGHT BC OF SUBDOMAIN
         if (x(A(ia))= xmax) then
                 id3 = id3+1
           A3(id3,1) = A(ia)
         end
# THE COORDINATES OF THE TOP BC OF SUBDOMAIN
         if (y(A(ia))= ymax & x(A(ia))>xmin & x(A(ia))<xmax) then
                 id2 = id2+1
           A2(id2,1) = A(ia)
         end

# THE COORDINATES OF THE BOTTOM BC OF SUBDOMAIN
         if (y(A(ia))= ymin & x(A(ia))>xmin & x(A(ia))<xmax) then
                 id4 = id4+1
           A4(id4,1) = A(ia)
         end
     end
#==================================================================
=====
```

```
# SET THE COORDINATES OF THE BC OF SUBDOMAIN AT THE BOTTOM DOMAIN
   elseif (j = 1) then
    if (i>1 & i<ndX) then
       id1 = 0
       id2 = 0
       id3 = 0
       id4 = 0
      for ia = 1 to nA(k)
        if (x(A(ia))= xmin & y(A(ia))>ymin) then
                 id1 = id1+1
           A1(id1,1) = A(ia)
        end
        if (x(A(ia))= xmax & y(A(ia))>ymin) then
                id3 = id3+1
          A3(id3,1) = A(ia)
        end
        if (y(A(ia))= ymax & x(A(ia))>xmin & x(A(ia))<xmax) then
                 id2 = id2+1
           A2(id2,1) = A(ia)
        end

        if (y(A(ia))= ymin) then

                      id4 = id4+1
                A4(id4,1) = A(ia)
            end
          end
        end
    #=================================================================
    =====


    # SET THE COORDINATES OF THE BC OF SUBDOMAIN AT THE LEFT-BOTTOM
    DOMAIN
     if (i = 1) then
            id1 = 0
            id2 = 0
            id3 = 0
            id4 = 0
        for ia = 1 to nA(k)
            if (x(A(ia))= xmin) then
                     id1 = id1+1
               A1(id1,1) = A(ia)
            end
            if (x(A(ia))= xmax & y(A(ia))> ymin) then
                    id3 = id3+1
              A3(id3,1) = A(ia)
            end
            if (y(A(ia))= ymax & x(A(ia))>xmin & x(A(ia))<xmax) then
                     id2 = id2+1
               A2(id2,1) = A(ia)
            end
            if (y(A(ia))= ymin & x(A(ia))>xmin) then
                     id4 = id4+1
               A4(id4,1) = A(ia)
            end
        end
       end
    #=================================================================
    ======
```

```
# SET THE COORDINATES OF THE BC OF SUBDOMAIN AT THE RIGHT-BOTTOM
DOMAIN
 if (i = ndX) then
        id1 = 0
        id2 = 0
        id3 = 0
        id4 = 0
     for ia = 1 to nA(k)
         if (x(A(ia))= xmin & y(A(ia))> ymin) then
                  id1 = id1+1
            A1(id1,1) = A(ia)
         end
         if  (x(A(ia))= xmax) then
                 id3 = id3+1
           A3(id3,1) = A(ia)
         end
         if (y(A(ia))= ymax & x(A(ia))>xmin & x(A(ia))<xmax) then
                  id2 = id2+1
            A2(id2,1) = A(ia)
            end
            if (y(A(ia))= ymin & x(A(ia))<xmax) then
                     id4 = id4+1
               A4(id4,1) = A(ia)
            end
        end

    end
  #=================================================================
  ======
   # SET THE COORDINATES OF THE BC OF SUBDOMAIN AT THE TOP DOMAIN
    elseif (j = ndY) then
        if (i>1 & i<ndX) then
          id1 = 0
          id2 = 0
          id3 = 0
          id4 = 0
       for ia = 1 to nA(k)
           if (x(A(ia))= xmin & y(A(ia))< ymax )then
                    id1 = id1+1
              A1(id1,1) = A(ia)
           end
           if (x(A(ia))= xmax & y(A(ia))< ymax ) then
                   id3 = id3+1
             A3(id3,1) = A(ia)
           end
           if (y(A(ia)) = ymax ) then
                    id2 = id2+1
              A2(id2,1) = A(ia)
           end
           if (y(A(ia)) = ymin & x(A(ia))>xmin & x(A(ia))<xmax)then
                    id4 = id4+1
              A4(id4,1) = A(ia)
           end
       end
      end
  #=================================================================
  ======
```

```
   # SET THE COORDINATES OF THE BC OF SUBDOMAIN AT THE LEFT-TOP
   DOMAIN
       if (i = 1) then
           id1 = 0
           id2 = 0
           id3 = 0
           id4 = 0
        for (ia = 1:nA(k)) then
            if  (x(A(ia))= xmin) then
                     id1 = id1+1
               A1(id1,1) = A(ia)
            end
            if  (x(A(ia))= xmax & y(A(ia))< ymax )then
                    id3 = id3+1
              A3(id3,1) = A(ia)
            end
         if  (y(A(ia))= ymax & x(A(ia))>xmin)then
                  id2 = id2+1
            A2(id2,1) = A(ia)
         end
         if (y(A(ia))= ymin & x(A(ia))>xmin & x(A(ia))<xmax) then
                  id4 = id4+1
            A4(id4,1) = A(ia)
         end
     end
    end
#=================================================================
======
  # SET THE COORDINATES OF THE BC OF SUBDOMAIN AT THE RIGHT-TOP
DOMAIN
     if (i = ndX) then
        id1 = 0
        id2 = 0
        id3 = 0
        id4 = 0
      for ia = 1 to nA(k)
         if  (x(A(ia))= xmin & y(A(ia))< ymax ) then
                  id1 = id1+1
            A1(id1,1) = A(ia)
         end
         if  (x(A(ia)) = xmax) then
                  id3 = id3+1
            A3(id3,1) = A(ia)
         end
         if  (y(A(ia)) = ymax & x(A(ia))<xmax) then
                  id2 = id2+1
            A2(id2,1) = A(ia)
         end
         if (y(A(ia)) = ymin & x(A(ia))>xmin & x(A(ia))<xmax) then
                  id4 = id4+1
           A4(id4,1) = A(ia)
         end
      end
  end
#=================================================================
=========
 end      (end if j>1 & j<ndY)
#=================================================================
=========
```

```
    n1 = id1
    n2 = n1+id2
    n3 = n2+id3
    n4 = n3+id4
# SET THE COORDINATES OF THE INNER SUBDOMAIN
     id5 = 0
  for ia = 1 to nA(k)
   if (x(A(ia))>xmin & x(A(ia))<xmax & y(A(ia))>ymin &
y(A(ia))<ymax)then
            id5 = id5+1
      Ai(id5,1) = A(ia)
    end
   end
      ni = id5
   for ia = 1 to nA(k)
      if (ia<= n1) then
          AA(ia) = A1(ia)
      elseif (ia>n1 & ia <= n2) then
          AA(ia) = A2(ia-n1)
      elseif (ia>n2 & ia <= n3) then
          AA(ia) = A3(ia-n2)
      elseif (ia>n3 & ia <= n4) then
          AA(ia) = A4(ia-n3)
      elseif (ia>n4 ) then
          AA(ia) = Ai(ia-n4)
      end
   end
   RETURN
```

3. SUB PROGRAM **RBF_DERIVATIVES_LU**

```
(Dx,Dy,Dxx,Dyy,DD) = RBF_Derivatives_LU(N,xA,yA,sp,k)
for I = 1 to N
   for j = 1 to N
     rx(i,j) = xA(i)-xA(j)
     ry(i,j) = yA(i)-yA(j)
      r(i,j) = sqrt(rx(i,j)^2+ry(i,j)^2)              (refer to
eq. 4)
      B(i,j )= sqrt(r(i,j)^2+sp^2)                    (refer to
eq. 6)
    Hx(i,j)  = rx(i,j)/sqrt(r(i,j)^2+sp^2)            (refer to
eq. 13)
    Hy(i,j)  = ry(i,j)/sqrt(r(i,j)^2+sp^2)            (refer to
eq. 13)
    Hxx(i,j) = (ry(i,j)^2+sp^2)/(r(i,j)^2+sp^2)^1.5   (refer to
eq. 21)
    Hyy(i,j) = (rx(i,j)^2+sp^2)/(r(i,j)^2+sp^2)^1.5   (refer to
eq. 21)
   end
end
CALL (invB) = LU_INVERSE(B,N)             (calculate the inverse
of B)
CALL   (Dx) = CROSS(Hx,invB,N,N,N)              (refer to eq. 16)
CALL   (Dy) = CROSS(Hy,invB,N,N,N)              (refer to eq. 16)
CALL  (Dxx) = CROSS(Hxx,invB,N,N,N)             (refer to eq. 23)
CALL  (Dyy) = CROSS(Hyy,invB,N,N,N)             (refer to eq. 23)

for i = 1 to N
   for j = 1 to N
     DD(i,j) = Dxx(i,j)+Dyy(i,j)
   end
end
RETURN
```

4. SUB PROGRAM **LU_INVERSE**

```
  (X) = LU_INVERSE(A,N)
CALL (L,U) = LU_DECOMPOSITION(A,N)
for k = 1 to N
     Z(k) = 1.0

 D(1,k) = Z(1)
     for i = 2 to N
         s = 0
         for j = 1 to i-1
             s = s+L(i,j)*D(j,k)
         end
         D(i,k) = Z(i)-s                          (refer to
eq. 42)
     end
      X(N,k) = D(N,k)/U(N,N)
     for i = N-1 to 1
         s = 0
         for j = N to i+1
             s = s+U(i,j)*X(j,k)
         end
        X(i,k) = (D(i,k)-s)/U(i,i)                (refer to
eq. 45)
     end
   Z(k)=0.
end
RETURN
```

5. SUB PROGRAM **LU_DECOMPOSITION**

```
(L,U) = LU_DECOMPOSITION(A,n)
for i = 1 to N
# Finding L
   for j = 1 to i-1
      L(i,j) = A(i,j)
    for k = 1 to j-1
      L(i,j) = L(i,j)-U(k,j)*L(i,k)               (refer to
eq. 39)
     end
      L(i,j) = L(i,j)/U(j,j)
   end

# Finding U
  for j = i to N
      U(i,j) = A(i,j)
   for k = 1 to i-1
      U(i,j) = U(i,j)-U(k,j)*L(i,j)               (refer to
eq. 38)

   end
  end
end

for i = 1 to N
  L(i,i) = 1
end
RETURN
```

6. SUB PROGRAM **CROSS**

```
(C) = CROSS(A,B,NA,MA,NB)
for i=1 to NA
    for k=1 to NB
        C (i,k)=0
        for j=1 to MA
         C(i,k)=C(i,k)+A(i,j)*B(j,k)
        end
    end
end
RETURN
```

```
7. SUB PROGRAM DOT
(C) = DOT(A,B,N)
  for I = 1 to N
     C(i)=A(i)*B(i)
  end
RETURN

8. SUB PROGRAM DOT_ROW
(C) = DOT_ROW(A,B,N)
for i = 1 to N
  for j = 1 to N
   C(i,j)= A(i)*B(i,j)
  end
end
RETURN

9. SUB PROGRAM IDENTITY
(A) = IDENTITY(N)
for i=1 to N
  for j=1 to N
    A(i,j) = 0
    A(i,i) = 1
  end
end
RETURN

10. SUB PROGRAM LU_SOLVE
(x) = LU_SOLVE(a,z,n)
CALL (L,U) = LU_DECOMPOSITION(a,n)
# Forward substitution to solve [L]{d} = {z}
 d(1) = z(1)
for i = 2 to n
    s = 0
  for j = 1:i-1
    s = s + L(i,j) * d(j)
  end
 d(i) = z(i) - s                                      (refer to
eq. 42)
end
# Back substitution to solve [U]{x} = {d}
 x(n) = d(n)/U(n,n)
for i = n-1 to 1
    s = 0
  for j = i+1:n
     s = s + U(i,j)*x(j)
  end

  x(i) = (d(i)-s)/U(i,i)                              (refer to
eq. 45)
end
RETURN
```

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